

Autonomous Boat Craft for Emergency Naval Delivery

A Major Qualifying Project submitted to the faculty of the Worcester Polytechnic Institute in partial fulfillment of the requirements for the Degree of Bachelor of Science in Mechanical Engineering

Submitted by:

Marcus Adams, Fabio Bastida Andreu, Andres Casanova-Aleman, Giovanni Crema, Samuel Kelly

Submitted to:

Mechanical Engineering Department of Worcester Polytechnic Institute

Advisor:

Selçuk Güçeri

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Abstract

Coastal natural disasters destroy port infrastructure, stranding supplies and leaving populations unreachable. Manned boats, drones, and ocean-bound ASVs lack the safety or capacity for near-shore delivery. This project addresses this gap by designing a fully autonomous, solar-powered monohull for emergency coastal supply. The fiberglass vessel features a 14.8V 20Ah LiPo system with dual 12V solar charging and a redundant BMS. Controlled by a Raspberry Pi 5 under ROS 2 Humble, it integrates GPS-IMU fusion, waypoint navigation, YOLO-based obstacle detection, and leak-sensing failsafes. To date, the system has successfully executed autonomous waypoint missions in controlled environments, with open-water validation ongoing.

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1.0 Introduction

1.1 The Problem

The frequency of reported natural disasters has increased significantly over the past 40 years. From 1995 to 2022, 11,360 natural disasters occurred, which killed 1,681,173 people and injured 7,599,259 others (Tin, D. et al. 2024). As the availability of technology changes in the world, we learn about how these natural disasters challenge humanitarian logistics. In the aftermath of events particularly affecting those in coastal regions, the necessity for fast medical intervention and supply delivery is lost due to the destruction of transportation infrastructure. While we have seen aerial solutions such as helicopters and drones being used, they are often constrained by weather conditions and load capacity.

Ports have become important nodes in humanitarian supply chain, having much of the vulnerable population reliant on last-mile distribution. Maritime Humanitarian Logistics have shown that ports are the first infrastructure to fail during coastal disasters. For example, Typhoon Haiyan in 2013, where a 5-meter storm surge destroyed port facilities in Tacloban, which had the region's main infrastructure inoperable and delayed relief efforts by days. Similarly, the 2010 Haiti Earthquake left only a single port operational for aid after it destroyed the main seaport's pier. In these cases, last-mile delivery remains hindered by the damaged infrastructure (Gavalas, D. et al., 2016).



Figure 1: Tsunami flooding in a coastal community.

This time in which the need for relief efforts and actual support is delivered creates a need for near-shore operations. Currently, large vessels are forced to find a solution to this issue

as shown in the aftermath of Cyclone Sidr in Bangladesh, where a U.S. Navy Destroyer was forced to hold station 25 miles offshore due to shallow coastal waters. Without an onboard helicopter or shallow-water landing craft, the ship "was unable to provide tangible relief to the devastated area" despite being on the scene (Yoho, K. et al. 2013).

As natural disasters become more complex and time important, private sector capabilities in innovation, global outreach, and quick deployment are increasingly useful for humanitarian operations. An article analyzing port performance during emergency relief efforts mentions that human safety is a major concern when sending rescue workers into debris-filled waters or storm surges (Gavalas, D. et al. 2026). Manned rescue vessels expose crews to significant risks from submerged debris and contamination, particularly at the initial response to a crisis.

As a result, last-mile logistics failures happen as supplies sit on large ships offshore or in distant operational ports, while humanitarian assistance is unreachable just a few miles away due to flooded roads and ruined docks. This creates a need for autonomous maritime vessels capable of navigating shallow, debris-filled waters to bridge this gap between deep-water relief vessels and the shoreline.

1.2 Existing Solutions

A variety of platforms currently exist for maritime, rescue, delivery and surveillance operations; however, each presents limitations when applied to autonomous emergency delivery in near-shore environments.

1.2.1 Human-Operated Rescue Boat

Human-operated rescue boats remain the most common solution for near-shore emergency response due to their high payload capacity, adaptability, and real-time decision-making capability (U.S. Coast Guard, 2022). These vessels are typically powered by gasoline or diesel engines and can operate in rough water conditions while transporting personnel and equipment. Despite their effectiveness, such systems rely heavily on trained operators, expose human crews to hazardous conditions, and incur high operational costs associated with fuel, staffing, and maintenance (International Maritime Organization, 2020). Additionally, response times can be limited by crew availability and deployment logistics, reducing effectiveness in time-critical emergency scenarios (U.S. Coast Guard, 2022).

1.2.2 Aerial Drones

Unmanned aerial vehicles (UAV's), commonly referred to as drones, have been explored for emergency delivery applications due to their rapid deployment and autonomy capabilities (Floreano & Wood, 2015). Drones are effective for short-range deliveries and reconnaissance; however, their payload capacity is typically limited to a few kilograms, constraining their usefulness for emergency supply delivery (Shakhathreh et al., 2019). Furthermore, UAV performance degrades significantly in high wind, rain, or maritime environments, which are common near shorelines (Dorling et al., 2017). Regulatory constraints, including airspace

restrictions and line-of-sight requirements, further limit their deployment in real-world emergency operations (Federal Aviation Administration, 2023). As a result, drones are not well-suited for sustained or reliable near-shore emergency delivery missions.

1.2.3 Autonomous Surface Vessels

Autonomous surface vessels (ASVs) have been widely researched for applications such as environmental monitoring, oceanographic research, and maritime surveillance (Manley, 2008). Commercial examples include long-endurance solar-powered platforms designed for open-ocean data collection, such as autonomous sailboats and wave-powered vehicles (Hine et al., 2009). While these systems demonstrate high endurance and autonomy, they are typically optimized for low-speed operation, minimal payloads, and calm or open-water conditions rather than dynamic near-shore environments (Manley, 2008). Additionally, many ASVs are designed primarily for data collection rather than targeted delivery tasks, lacking mechanisms for precise navigation to coastal locations or delivery of physical payloads (Bertram, 2012).

1.2.4 Gap in Existing Solutions

Despite the availability of human-operated rescue boats, aerial drones, and autonomous surface vessels, none are specifically optimized for autonomous emergency delivery in near-shore environments. Human-operated vessels expose crews to risk and require significant resources, drones suffer from payload and environmental limitations, and existing ASVs prioritize endurance and data collection over rapid, reliable delivery (Manley, 2008; Dorling et al., 2017). This gap highlights the need for a dedicated autonomous surface platform capable of operating in rough near-shore conditions, carrying meaningful emergency payloads, and functioning without direct human intervention.

1.3 Proposed Solution

This project proposes the development of a solar-powered and fully autonomous monohull surface vessel designed for near-shore emergency delivery. The surface vessel utilizes multiple GPS, camera, and water sensors for precise navigation, capable of being paired with supply chain management software to efficiently allocate and coordinate the distribution of supplies. This integration of logistics planning and autonomous control enables the vessel to deliver emergency packages safely, repeatedly, and exactly where they are needed.

1.4 Mission Statement & Objectives

This Major Qualifying Project's mission statement and objectives are discussed below.

1.4.1 Mission Statement

The mission behind this project is to design, manufacture, and test a fully autonomous, solar-powered surface vessel capable of navigating rough marine environments to safely deliver emergency supplies. The system aims to reduce human risk and response time in coastal disaster

scenarios by utilizing a monohull platform integrated with renewable energy harvesting and autonomous waypoint navigation.

1.4.2 Objectives

To achieve this mission, the team established the following specific technical and operational objectives:

- **Analyze Existing Maritime Solutions:** Conduct a review of current autonomous surface vessels, hull hydrodynamics, and solar-marine systems to identify design requirements.
- **Design & Fabricate a Monohull Platform:** Engineer a monohull structure optimized for positive buoyancy and reduced drag to ensure vessel stability while protecting internal electronics.
- **Develop a Sustainable Energy Model:** Create a power budget and integrate a solar charging system capable of sustaining the propulsion and onboard electronics for extended mission durations.
- **Implement Autonomous Navigation:** Code an autonomy stack utilizing GPS and sensor telemetry to execute end-to-end missions without real-time human intervention.
- **Integrate Sensor Instrumentation:** Incorporate sensors to provide real-time environmental awareness and system health monitoring.
- **Validate System through Field Testing:** Verify the system's capabilities with structured testing in a controlled aquatic environment.

2.0 Background

2.1 Emergency Delivery in Marine Environments

Emergency delivery in marine and coastal environments refers to the rapid transportation of critical supplies such as medical equipment, food, and rescue tools to affected populations following natural disasters or infrastructure failures. Unlike traditional supply chains, emergency logistics operate under highly uncertain and time-constrained conditions, where infrastructure is often damaged, communication systems are disrupted, and access routes are limited (Gavalas et al., 2026; Van Wassenhove, 2006). In coastal regions, these challenges are intensified due to the dependence on maritime transportation and the vulnerability of ports and shoreline infrastructure.

This section introduces the concept of maritime humanitarian logistics, examines the challenges associated with coastal delivery, and discusses the time-sensitive and high-risk nature of emergency response operations.

2.1.1 Maritime Humanitarian Logistics (MHL)

Maritime Humanitarian Logistics (MHL) describes the planning and execution of relief operations that rely on sea-based transportation systems to deliver aid to disaster-affected regions. In contrast to commercial logistics, which prioritize efficiency and cost minimization, MHL focuses on responsiveness, flexibility, and the ability to operate under uncertain conditions (Yoho et al., 2013; Van Wassenhove, 2006).

Ports play a central role in these operations, acting as the primary entry points for humanitarian aid. However, they are also highly vulnerable to damage during coastal disasters. Disruptions to port infrastructure significantly reduce the efficiency of relief efforts, as vessels carrying supplies may be unable to dock or unload cargo (Gavalas et al., 2026). When port capacity is compromised, supply chains become fragmented, and alternative delivery methods must be used, often resulting in delays and reduced effectiveness.

In addition, maritime logistics systems must function in environments where coordination mechanisms and transportation networks are partially or completely disabled. This creates challenges in organizing distribution and increases the difficulty of ensuring that supplies reach their intended destinations (Liang et al., 2025). These conditions highlight the importance of decentralized and adaptable delivery systems that can operate independently of traditional infrastructure.

2.1.2 Challenges in Coastal Logistics

Coastal disaster environments present unique logistical challenges that differ from inland scenarios. One of the primary issues is the loss or degradation of infrastructure, including ports, roads, and docking facilities, which are essential for transferring supplies from large vessels to affected populations (Gavalas et al., 2026). When these systems fail, aid may remain offshore or in distant operational hubs, creating a gap between supply availability and demand.

Near-shore regions further complicate delivery operations due to shallow waters, submerged obstacles, and floating debris. Large vessels are typically unable to operate safely in these areas, while smaller manned boats face increased risks associated with unstable water conditions and limited maneuverability (Yoho et al., 2013). This creates a significant challenge in completing last-mile delivery from offshore supply points to the shoreline.

Environmental conditions such as flooding and storm surges also alter terrain and introduce dynamic obstacles that hinder navigation. In densely populated coastal areas, these factors can lead to congested waterways and reduced operational efficiency (Liang et al., 2025). As a result, effective coastal logistics require systems capable of navigating shallow, debris-filled, and continuously changing environments.

2.1.3 Time Sensitivity in Emergency Response

Time is a critical factor in disaster response, as delays in delivering essential supplies can significantly increase casualties and worsen the overall impact of the event. Emergency logistics systems are therefore designed to prioritize rapid response and efficient distribution under constrained conditions (Yoho et al., 2013; Beamon & Balcik, 2008). In many cases, there is a limited time window during which intervention is most effective in preventing loss of life and stabilizing affected populations. However, logistical bottlenecks such as damaged infrastructure, limited transportation assets, and coordination delays often prevent timely delivery (Gavalas et al., 2026). Maritime operations are particularly affected, as vessels may be required to remain offshore until safe access to coastal areas can be established.

The inability to perform efficient last-mile delivery in near-shore environments is a major limitation of current systems. Even when supplies are available, delays in transporting them from offshore vessels to the shoreline reduce their effectiveness and can lead to shortages of critical resources (Liang et al., 2025). Improving delivery speed and reliability in these environments is therefore a key objective in modern humanitarian logistics.

2.1.4 Operational Risks in Maritime Emergency Delivery

Emergency delivery operations in marine environments involve significant risks for both personnel and equipment. Manned rescue missions often require crews to operate in hazardous conditions, including strong currents, debris-filled waters, and contamination from hazardous materials released during disasters (Gavalas et al., 2026). These risks are especially pronounced during the initial stages of response, when environmental conditions are unstable and situational awareness is limited.

In addition to human safety concerns, system reliability is a major challenge in marine environments. Saltwater exposure accelerates corrosion, while continuous wave impacts introduce mechanical stress that can degrade structural components over time. Water ingress also poses a serious threat to onboard electronics, potentially leading to sensor failure or complete system shutdown (Jorge et al., 2019). Although autonomous systems can reduce human exposure

to these risks, most existing unmanned surface vehicles are not optimized for near-shore emergency delivery. They often lack sufficient payload capacity, maneuverability, and robustness to operate effectively in harsh and unpredictable conditions (Jorge et al., 2019).

2.1.5 Implications for Autonomous Marine Delivery Systems

The challenges associated with maritime emergency delivery highlight a clear gap between offshore supply availability and onshore demand. Existing systems are often unable to complete the final stage of delivery efficiently, particularly in near-shore environments where infrastructure is damaged and access is limited.

Traditional approaches rely heavily on manned vessels, which are not always suitable for shallow or debris-filled waters. This creates delays and increases operational risk, especially in time-critical scenarios where rapid response is essential.

Autonomous surface vessels offer a potential solution by enabling delivery operations without direct human involvement. Their ability to operate continuously and navigate constrained environments makes them well suited for near-shore applications.

This project builds on these requirements by proposing a system specifically designed for emergency delivery in coastal environments. The focus is on achieving a balance between payload capacity, maneuverability, and reliability, ensuring that the vessel can perform effectively under realistic disaster conditions.

2.2 Near-Shore Ocean Environmental Challenges

Operating an autonomous surface vessel in the littoral (near-shore zone) presents a set of dynamic and static variables distinct from deep-ocean environments. Although wave energy is often consistent in open waters, near-shore environments are characterized by quick bathymetric changes and highly corrosive elements. This section analyzes the main environmental factors that influence the system design of the emergency delivery craft.

2.2.1 Hydrodynamic and Aerodynamic Forces

The near-shore environment subjects the vessel to interacting forces from waves, wind, and currents. Specifically, as deep-water waves approach the shoreline, they interact with the seabed, causing them to slow down and increase in steepness. This is a phenomenon referred to as shoaling. For a small-scale monohull, this results in high-frequency pitch and roll movements that can destabilize the platform and disrupt sensor telemetry (Abdelaal et al., 2018).

Additionally, near-shore currents can exceed the cruising speed of low-powered electric propulsion systems due to tidal fluxes or local wind patterns. The vessel's propulsion system must be sized to overcome these opposing drag forces while maintaining a heading. Wind loads on the vessel's freeboard can create significant yaw moments, requiring active course correction by autonomy algorithms.

2.2.2 Navigational Hazards and Obstruction

Navigation in the littoral zone requires higher precision than open-ocean transit due to the density of hazards. Static hazards refer to shallow drafts, submerged rocks, sandbars, and man-made structures that require the vessel to maintain strict adherence to defined waypoints. Aside from said hazards, the vessel has the risk of fouling the propeller or damaging the hull due to floating debris and biological obstacles. These are referred to as dynamic hazards. In addition, near-shore areas are often populated with recreational and commercial vessels, requiring powerful collision avoidance protocols and strict geofencing to prevent accidents.

2.2.3 GPS Signal Degradation

While Global Positioning Systems (GPS) are the common method for autonomous localization, near-shore environments are prone to multipath interference. This occurs when satellite signals bounce off vertical surfaces before reaching the receiver, resulting in timing errors that can harm positional accuracy by several meters (Zhang et al., 2022).

For an autonomous delivery mission that requires precise docking and landing, standard GPS accuracy could be insufficient. This challenge reflects the importance of implementing sensor fusion and potentially using an Inertial Measurement Unit (IMU) or compass. Said implementations would help filter noisy GPS data and maintain functioning capabilities during momentary signal loss.

2.2.4 Corrosive Environment and Ingress Protection

Saltwater is a highly aggressive electrolyte that accelerates corrosion, specifically through galvanic action where dissimilar metals are in electrical contact (Fossen, 2021). All exposed metallic components must be selected from marine-grade materials to prevent structural failure.

Furthermore, the integration of sensitive electronics requires rigorous waterproofing. The vessel's hull and electronics enclosure must meet high Ingress Protection standards to withstand submersion and the high-pressure impact of breaking waves. Additionally, heat dissipation becomes a challenge given that sealed enclosures trap heat generated by electronics. This requires careful thermal management design.

2.3 Solar Power for Autonomous Marine Robotics

Solar energy has emerged as a promising power source for autonomous marine robotics as researchers seek alternatives to fuel-based and purely battery-driven systems. This section reviews the motivation for adopting solar power in marine platforms, summarizes demonstrated applications, and discusses key advantages and limitations associated with solar-powered operation in maritime environments.

2.3.1 Motivation for Solar Energy in Marine Robotics

Solar energy has become an increasingly attractive power source for autonomous marine robotic systems due to its ability to provide long-duration operation without reliance on refueling or frequent battery replacement (Hine et al., 2009). In contrast to conventional internal combustion or purely battery-powered platforms, solar-powered vessels can harvest renewable energy during operation, enabling extended missions while reducing operational cost and environmental impact (Manley, 2008). This capability is particularly valuable for autonomous surface vessels intended for persistent monitoring, data collection, or delivery tasks, where endurance and reliability are critical performance metrics (Bertram, 2012).

2.3.2 Demonstrated Use of Solar-Powered Autonomous Vessels

Several autonomous marine platforms have demonstrated the feasibility of solar-powered propulsion and onboard electronics. Long-endurance autonomous boats have successfully operated for weeks to months using deck-mounted photovoltaic panels combined with battery storage systems (Hine et al., 2009). These systems leverage high-efficiency solar cells and power management electronics to sustain low-speed propulsion, navigation sensors, and computing hardware (Manley, 2008). The absence of fuel consumption significantly lowers recurring operational expenses and simplifies logistics, making solar-powered platforms attractive for applications requiring frequent or prolonged deployment (Bertram, 2012).

2.3.3 System Scalability, Quiet Operation, and Environmental Constraints of Solar Power

Beyond endurance advantages, solar power also supports modular and scalable system architectures for autonomous marine platforms. Power generation capacity can be increased by adjusting panel area, while energy storage capacity can be modified through battery selection, allowing designers to tailor systems to specific mission requirements (Hine et al., 2009). In addition, electric propulsion systems powered by solar-charged batteries operate relatively quietly, which is advantageous for missions near populated shorelines and in environmentally sensitive regions (Manley, 2008). However, these benefits are accompanied by important limitations. Solar energy availability is highly dependent on environmental conditions such as cloud cover, time of day, and seasonal variation, resulting in fluctuating power generation (Messenger & Ventre, 2010). This variability can restrict mission duration or necessitate conservative power budgeting strategies to ensure continuous operation during periods of low irradiance.

2.3.5 Deck Area, Power Density and Battery Limitations

Physical constraints further limit solar integration on small surface vessels. The available deck area directly bounds the maximum achievable solar panel size, thereby capping total power generation (Hine et al., 2009). As a result, most solar-powered autonomous boats operate at relatively low speeds and carry lightweight payloads to maintain energy balance (Manley, 2008). For missions requiring higher propulsion power or heavier payloads, energy storage systems must compensate for insufficient solar input, increasing system mass and volume.

Battery energy density also constrains solar-powered designs. Although lithium-based batteries offer high energy density compared to other rechargeable chemistries, their stored energy remains limited relative to liquid fuels, which can restrict peak power delivery and mission duration under high-load conditions (Nitta et al., 2015). Charging time represents an additional limitation, as replenishing depleted batteries through solar input alone may require several hours of favorable sunlight conditions (Messenger & Ventre, 2010).

2.3.7 Implications for Near-Shore Emergency Delivery

Consequently, most successful solar-powered autonomous marine platforms prioritize energy-efficient operation, lightweight structures, and low-speed propulsion to maintain feasible energy budgets (Hine et al., 2009). For near-shore emergency delivery applications, solar power is therefore best utilized as a supplementary or primary energy source supporting efficient electric propulsion and onboard electronics, rather than as the sole determinant of system performance. Careful system-level power modeling and component selection are essential to ensure reliable operation under realistic environmental conditions.

2.4 Existing Robotic Solutions

The domain of unmanned maritime operations has expanded significantly, evolving from simple remote-controlled boats to sophisticated Autonomous Surface Vehicles (ASVs) capable of complex decision-making. Current solutions can be broadly categorized into commercial hydrographic survey vessels, long-endurance solar platforms, and smaller experimental delivery prototypes.

2.4.1 Autonomous Surface Vehicles (ASVs)

Commercially available ASVs are currently the industry standard for hydrographic and bathymetric surveys. Companies such as SeaRobotics have developed modular systems like the SR-Surveyor, which are designed to carry diverse payloads including multi-beam echo sounders and LiDAR (SeaRobotics, 2024). These platforms prioritize stability and payload integration over long-range endurance. However, the efficacy of these vessels relies heavily on advanced guidance systems. According to a recent review by Wang et al. (2023), modern ASVs are increasingly adopting hybrid path-planning algorithms that combine global GPS navigation with local reactive obstacle avoidance to navigate dynamic marine environments safely.

2.4.2 Solar Boats and Long-Endurance Systems

To address the range limitations of battery-only ASVs, solar-powered Unmanned Surface Vehicles (USVs) have emerged as a viable solution for continuous ocean monitoring. Open Ocean Robotics has pioneered this sector with the *Data Explorer*, a self-righting USV capable of remaining at sea for months without refueling (Open Ocean Robotics, 2024). These systems rely on harvesting renewable energy to power both propulsion and onboard sensors. Recent studies in sensor integration highlight that the challenge for these vessels is not just power generation, but

intelligent energy management; systems must dynamically allocate power between thrust and payload data transmission based on environmental conditions (MDPI, 2023).

2.4.3 Small-Scale Delivery and Prototype Robots

While large industrial vessels dominate the survey market, smaller robotic platforms are being developed for "last-mile" logistics and academic research. Unlike aerial drones, water-based delivery robots must contend with significant hydrodynamic drag and wave perturbations. Constant (2020) demonstrated the feasibility of low-cost autonomous boats using off-the-shelf microcontrollers and GPS modules. This research highlights that while small-scale prototypes often lack the range of commercial surveyors, they serve as critical testbeds for developing the swarming logic and point-to-point navigation algorithms required for future autonomous delivery networks (Constant, 2020).

2.4.4 Summary of Technology Gap

Reviewing these benchmarks reveals a distinct gap in the current market. High-end systems like the *Data Explorer* or *SR-Surveyor* offer exceptional autonomy and range but come with high costs and complexity prohibitive for smaller-scale applications. Conversely, academic prototypes (Constant, 2020) are cost-effective but often lack the seaworthiness or power management required for practical deployment. The proposed design aims to bridge this gap by balancing the modularity of commercial survey boats with the accessible architecture of academic prototypes.

3.0 Design Specifications

The design specifications for the Autonomous Boat Craft for Emergency Naval Delivery were developed based on the application and described in the: a small, unmanned surface vessel capable of transporting emergency supplies through shallow, near-shore marine environments where current systems would be considered unapplicable. Autonomous surface vessels used in disaster response must have solid payload capacity, stability, maneuverability, sensing, and reliability under uncertain environmental conditions (Jorge et al., 2019). In addition, marine craft design requires that buoyancy, resistance, propulsion, and control requirements be thought of in design together, because changes in hull geometry, mass distribution, and payload directly affect stability and maneuverability (Fossen, 2021).

For this project, the specifications were written as proof-of-concept engineering requirements rather than full commercial deployment requirements. The vessel was not intended to replace large rescue craft or certified maritime emergency systems. Instead, it was designed to demonstrate that a compact, solar-assisted, autonomous monohull platform could carry a small emergency payload, remain positively buoyant, protect its internal systems, maneuver using a dual thrust differential, and complete waypoint-based delivery missions in controlled water testing. This approach is based off the role of small autonomous surface vehicle prototypes as testbeds for validating mission-specific navigation, payload, and platform design concepts before scaling to operational systems (Manley, 2008; Sornek et al., 2022).

These specifications served as the connection between the project objectives and the detailed subsystem designs described in the following chapters. Rather than optimizing the prototype for maximum speed, long-range endurance, or full open-ocean operation, the design prioritized a practical balance of flotation, payload support, waterproofing, maneuverability, serviceability, and autonomous mission execution. This was important because the craft was intended to demonstrate near-shore emergency delivery at prototype scale, where reliable operation and payload survivability were more important than high-speed travel or full deployment readiness.

3.1 System-Level Design Requirements

The primary design requirements are summarized in Table 1. These requirements define what the vessel needed to achieve, while later sections of the report describe the mechanical construction, electrical architecture, software implementation, manufacturing process, and testing results in greater detail.

Design Category	Specification	Target / Final Value	Purpose
Mission Application	Emergency near-shore supply delivery	Autonomous delivery of small emergency payloads	Addresses the gap between offshore supply availability and damaged shoreline access
Hull Configuration	Surface vessel hull type	Monohull	Provides a compact, manufacturable platform with sufficient internal volume
Overall Dimensions	Length $\times$ width $\times$ height	24 in $\times$ 12.7 in $\times$ 9 in	Establishes size limits for fabrication, transportability, and shallow-water operation
Fully Assembled Weight	Total vessel weight	19.3 lb	Defines the operating mass used for buoyancy and propulsion analysis
Payload Capacity	Emergency supply payload	6.0 lb	Enables transport of small medical, communication, or survival supplies
Displaced Volume	Maximum displaced volume	952.68 in ³	Determines maximum buoyant support of the hull
Maximum Buoyant Support	Maximum supported weight in freshwater	34.4 lb	Confirms that the hull can support the vessel and payload with reserve buoyancy
Reserve Buoyancy	Remaining buoyant support before payload	15.1 lb	Provides margin for payload, water disturbance, and small added components

Displacement Utilization	Vessel weight / maximum buoyant support	56.10%	Confirms that the craft does not operate near its flotation limit
Stability Requirement	Static flotation and trim	Maintain positive flotation and acceptable trim under payload	Ensures payload does not prevent vessel operation
Waterproofing Requirement	Hull water resistance	No internal water ingress during short-duration submersion/leak testing	Protects internal components and payload during operation
Autonomy Requirement	Mission execution	Complete waypoint-based navigation without continuous operator input	Demonstrates core autonomous delivery functionality
Navigation Accuracy	Delivery waypoint tolerance	Approximately 3 m after a 100 m waypoint mission	Provides a measurable target for controlled proof-of-concept validation
Sensor Integration	Environmental and system awareness	GPS, camera, Wi-Fi, water sensing, and onboard telemetry	Supports navigation, monitoring, and emergency response behavior
Field Validation	Required test categories	Float, leak, propulsion, steering, waypoint navigation, sensor integration, and solar integration tests	Confirms that the system functions as an integrated prototype

Table 1: Design requirements.

3.2 Buoyancy and Payload Specification

The most important physical design requirement was that the vessel remain positively buoyant while carrying its intended emergency payload. The vessel has a displaced volume of 952.68 in³, corresponding to a maximum buoyant support of approximately 34.4 lb. Because the assembled vessel weighs 19.3 lb, the unloaded craft uses approximately 56.1% of its available buoyant capacity. This leaves 15.1 lb of reserve buoyancy before payload.

The target payload capacity was set at 6.0 lb. With this payload included, the total loaded weight becomes 25.3 lb, giving an estimated loaded displacement utilization of approximately 73.5%. This means that the boat has roughly 9.1 lb of additional buoyant support after carrying payload. This reserve is important because a near-shore vessel cannot be designed to operate at the edge of neutral buoyancy. Waves, trim changes, water on deck, imperfect weight distribution, and added components can all reduce the effective flotation margin. Therefore, the design specification required the vessel to support the complete operating mass and payload while maintaining positive flotation and stable trim.

This buoyancy margin also supports the project's emergency delivery objective. A vessel that carries a payload but becomes unstable, excessively trimmed, or nearly submerged would not be suitable for autonomous operation, due to the nature of unmanned naval operation. As a result, payload capacity was not defined only by the available space inside the boat. It was defined by the relationship between hull displacement, total system weight, and acceptable stability during operation.

The loaded displacement condition was therefore the most important design case for the prototype. While the unloaded displacement utilization confirmed that the bare craft had significant flotation margin, the loaded condition better represented the actual mission scenario. With the 6.0 lb payload installed, the vessel operated at approximately 73.5% of its maximum buoyant support, leaving roughly 9.1 lb of remaining buoyancy. This remaining margin was necessary to account for small uncertainties in mass distribution, water accumulation on the hull surface, minor added components, and trim variation during motion.

3.3 Hull and Dimensional Specification

The hull was specified as a compact monohull with final dimensions of 24 in in length, 12.7 in in width, and 9 in in height. These dimensions were selected to create a balance between manufacturability, transportability, buoyancy, and internal packaging volume. A smaller hull improves portability and reduces material cost, but it also limits displacement volume, payload capacity, and available space for onboard systems. Conversely, a larger hull would increase buoyancy and internal space but would require greater propulsion force and more fabrication time.

The dimensional specification also reflected a serviceability requirement. Because the vessel had to carry a payload while also housing internal systems, the hull could not be treated as a permanently sealed shell. The design required a removable upper access structure so the team could load emergency supplies, inspect the internal compartment, and perform maintenance between tests. This access requirement influenced the overall hull height and internal volume, since the craft needed enough space for payload storage and subsystem packaging while still maintaining a compact external footprint.

The monohull configuration was selected for this proof-of-concept because it provided a continuous internal volume for payload and system integration while maintaining a simple hull geometry for manufacturing. In the context of autonomous surface vessels, hull geometry is the most important thing for payload displacement, dynamics, drag and buoyancy, making the hull form a central system-level specification rather than only a mechanical design detail (Fossen, 2021).

3.4 Propulsion and Maneuverability Specification

The vessel was designed to use dual electric thruster propulsion. At the design specification level, the propulsion system needed to satisfy two requirements: generate sufficient forward thrust to move the loaded vessel through water conditions and create differential thrust for steering. This eliminated the need for a separate rudder system and allowed turning to be controlled by varying the relative thrust of the left and right motors. Also enabling the ability to turn on the spot increasing maneuverability in tight spaces. At the specification level, the key requirement was not simply motor power, but the ability of the loaded craft to translate forward and rotate predictably at low speed.

The propulsion specification was intentionally tied to observable performance rather than only motor rating. The vessel needed to demonstrate stable forward motion while carrying its payload and controlled turning through differential thrust. This requirement reflects the operating reality of small autonomous surface vessels, where maneuverability depends not only on available motor power but also on hull resistance, weight distribution, environmental disturbance, and control response. Prior research on autonomous surface vessels emphasizes that successful design requires integration between platform dynamics, propulsion, and navigation control rather than treating propulsion as an isolated subsystem (Manley, 2008; Fossen, 2021).

3.5 Waterproofing and Environmental Protection Specification

Because the vessel is intended for near-shore marine operation, waterproofing was specified as a primary design requirement. The hull needed to protect the payload and onboard systems from water intrusion during normal operation, splashing, and short-duration exposure events. The validation target for this prototype was a controlled leak/submersion test, with the requirement that no internal water ingress occur during the test period.

This requirement is especially important for autonomous surface vessels because water ingress can ruin sensing, control, propulsion, and mission completion. Disaster-response USVs must operate in uncertain environments where debris, waves, and contamination can increase the risk of hull damage or internal exposure (Jorge et al., 2019). Therefore, waterproofing was treated as a system-level requirement rather than simply a manufacturing detail.

We also incorporated a waterproof housing inside of the vessel that stores all the electronics. From the electronics all wires run out of the top that is sealed using silicon sealant and marine grade epoxy. This is also used to run all the wires out of the boats top near where the

camera lies, sealed in the same fashion. The wires sealed include: 2 motors, 1 camera, 1 GPS, 2 solar panels, and 1 Wi-Fi antenna.

The design functions like a rooftop cargo box with the top completely removable to the bottom, allowing ease of access to the emergency payload and to the electronics. This is done with a marine grade rubber sealant tape separating the top of the vessel, top of the housing and the bottom of the boat and bottom on the housing. To compress and secure the vessel, rubber latches are screwed into the top and bottom. The rubber allows the latches to follow the complex curvature of the boat, enabling consistent pressure through the contact area. These are all then epoxy sealed again.

For the design specification stage, waterproofing was therefore defined by function rather than by the exact fabrication method. The craft needed to provide a protected internal environment while still allowing repeated access for loading, inspection, and repair. The detailed material selection, sealing process, and assembly method are addressed in the mechanical design and manufacturing sections.

3.6 Solar-Assisted Operation Specification

The vessel was specified as a solar-assisted electric platform. The purpose of the solar subsystem was to supplement onboard energy storage and demonstrate the feasibility of renewable power integration in a compact autonomous surface vessel. Solar-powered autonomous vessels are commonly limited by available deck area, solar irradiance, vehicle drag, and onboard power demand; therefore, solar integration must be evaluated as part of the overall energy balance rather than as a stand-alone feature (Sornek et al., 2022).

For this prototype, the design specification was not that the vessel operates indefinitely on solar power alone. Instead, the requirement was that the solar subsystem be physically integrated, function during system operation, and provide a basis for future quantitative endurance testing. This avoided overstating the capability of the prototype while still aligning with the project objective of developing a solar-powered emergency delivery platform.

4.0 Mechanical Design

4.1 Hull Selection & Geometry

The selection of a monohull over a multi-hull configuration (such as a catamaran) was an engineering decision based on the mission's requirement for internal payload volume and structural simplicity. While catamarans offer high initial stability and a reduce the wetted surface, they have limited internal space for an emergency payload. A monohull allows for a deep, centralized keel, which is essential for housing heavy power systems to lower the center of gravity.



Figure 2: Boat Terminology

Figure 1 shows commonly used nautical terms by boat designers. It is important to have an idea about these terms before going into detail (Varghese, 2010).

4.1.1 Chines

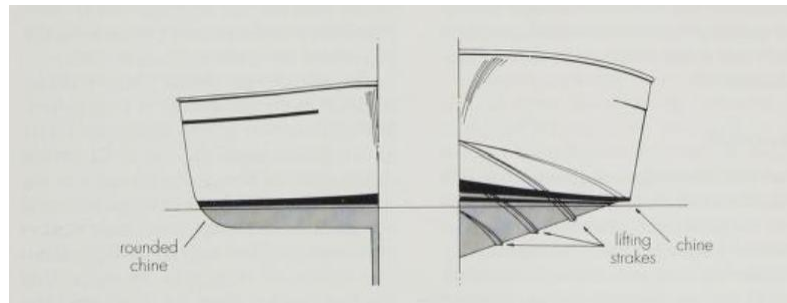


Figure 3: Comparison of chines on boat.

When looking at a boat outside of the water, you'll notice a sharp change where the side of the boat meets the hull bottom. There are 3 different types of chines, soft (rounded), hard (angular), or reverse.

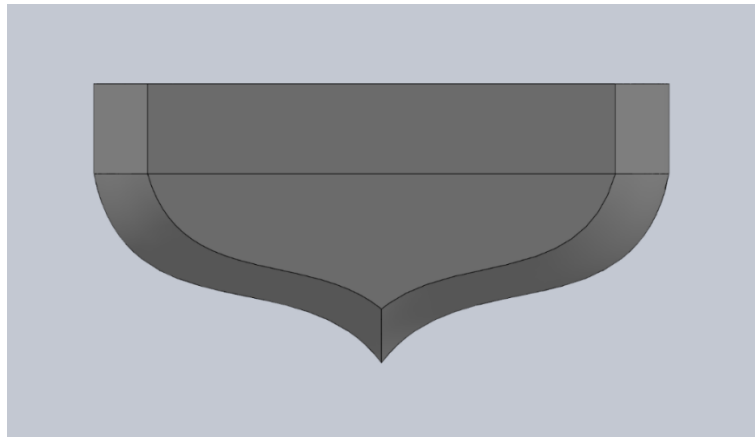


Figure 4: Back view of hull.

The hull features a rounded chine, characterized by a curved transition between the hull bottom and the sides. This design choice increases initial stability by resisting rolling moments and provides additional lift, which is critical for maintaining a level platform for the solar panels and camera mounted on the top of the boat (Lutfi et al. 2024).

4.1.2 Hull Type

The vessel utilizes a semi-displacement hull form to balance having payload capacity with speed requirements. Unlike a fully displacement hull, which is physically limited by its speed, our semi-displacement form uses a modified-V shape to generate partial hydrodynamic lift when it is in motion. This reduces the wetted surface at higher speeds, which allows the vessel to overcome water resistance more efficiently. Additionally, the S-curved profile creates a narrow entry at the bow to minimize drag while widening at the midsection to provide the necessary volume for internal components (Marshall, 2002).

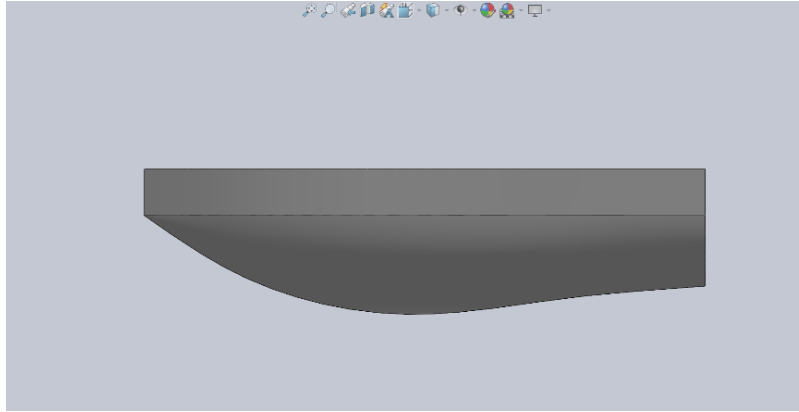


Figure 5: Right Plane view of hull.

4.1.4 Boat Stability, Center of Gravity and Buoyancy

A primary concern for autonomous coastal vessels is turtling or being flipped due to rough water conditions. The hull geometry is engineered to mitigate this risk through secondary stability. The rounded chines provide high initial stability (resistance to tipping in calm water). However, the S-curve gives it secondary stability. In other words, as the boat begins to plane, the rapidly increasing submerged volume of the sides generates a massive Righting Moment. This Righting Moment is the effect of placing the 14.8V battery and motors in the back part of the keel. This causes the Center of Gravity to be kept significantly lower than the Center of Buoyancy, returning the vessel to equilibrium. The boat naturally seeks its upright equilibrium, even when influenced by significant displacement in debris filled zones (Marshall, 2002).

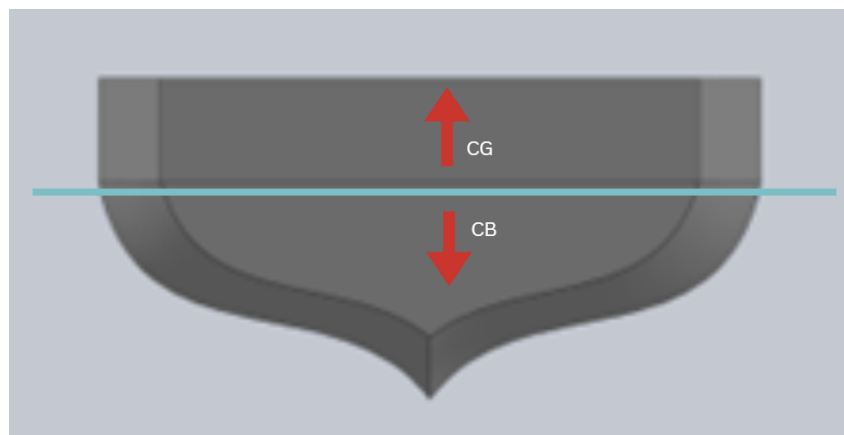


Figure 6: Center of gravity & buoyancy diagram in static condition.

Hydrostatic Calculations:

$$\text{Displacement Utilization} = (19.3/34.4) \times 100$$

$$\text{Displacement Utilization} = 56.1\%$$

- The vessel is designed to operate at approximately 56.1% displacement utilization under standard mission loads.

Parameter	Value
Length Overall	24.0 in
Beam	12.7 in
Total Hull Volume (V)	952.68 in ³
Dry Weight (W dry)	19.3 lbs
Max Buoyant Support	34.4 lbs
Optimized Payload	6.0 lbs
Reserve Buoyancy	15.1 lbs (~44%)

Table 2: Mechanical specs.

With a total hull volume of 952.68 in³ the maximum buoyant support is 34.4 lbs. At the operational weight of 25.3 lbs, the vessel maintains a reserve buoyancy of 26.5%, which is the critical safety margin required to prevent swamping during autonomous navigation in coastal swells.

4.1.3 Wet Surface and Draft

Using Archimedes' Principle, the vessel's draft (D) is calculated by balancing the total mass against the displaced volume of water.

Parameter	Symbol	Value
Total Operational Mass	m	25.3 lbs.
Length Overall	L	24 in.
Beam (Maximum Width)	W	12.7 in.
Density of Water	ρ	62.43 lb/ft ³
Block Coefficient	C_b	0.7

Table 3: Values and symbols for draft calculation.

$$m = \rho \cdot L \cdot W \cdot D \cdot C_b$$

$$\rightarrow D = 3.31 \text{ in.}$$

The hull's flat and wide mid-section increases the waterplane area, which helps with stability and reduces the draft to a calculated 3.31 inches. Given the hull height of 9 inches, this confirms a safe operational freeboard. This shallow draft is important for last-mile delivery missions. It allows the vessel to navigate over submerged obstacles which are close to the shoreline, which would be inaccessible to deeper keeled rescue boats. A round hull (Fig. 6) is often preferred for low-speed efficiency because it minimizes wetted surface area and hydrodynamic drag. However, this shape misses lateral resistance, making it prone to excessive rolling and swaying when attempting to maintain straight movement.

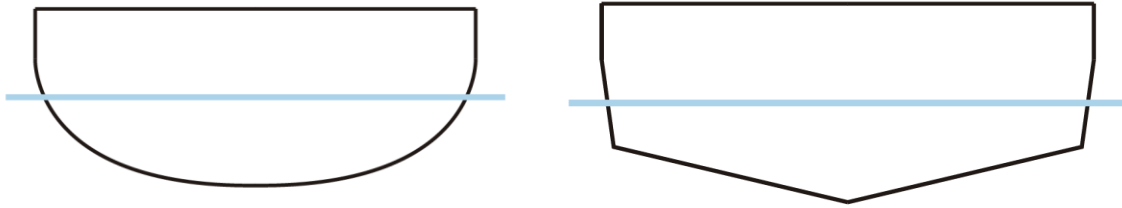


Figure 7: Shallow V hull. & Figure 8: Round hull.

In contrast, the shallow V hull (Fig. 7) incorporates a sharper keel-like geometry that significantly improves directional stability. This prevents our vessel from steering of course or excessive sway during high-speed autonomous missions (Lutfi et al. 2024).

4.1.5 Final Selection

For the vessel's hull selection, our design classifies as a S-curved semi-displacement hull. Its flat and wide structure will help with stability, and the shallow keel helps maintain directional stability in shallow water/coastal environments.

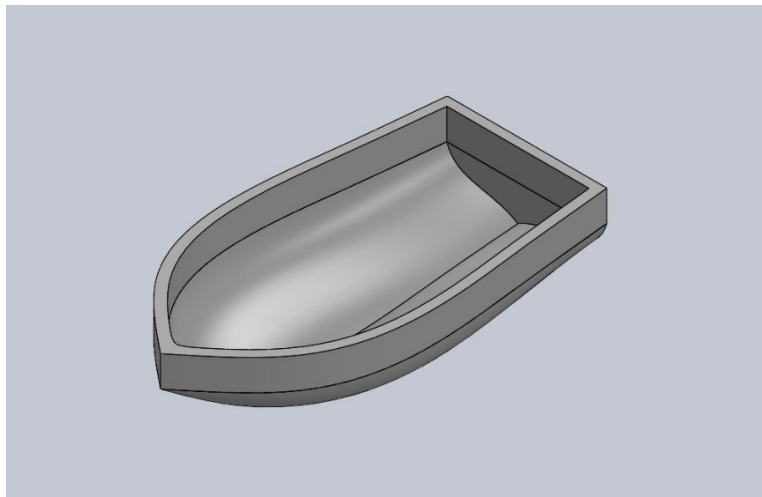


Figure 9: Isometric view of hull.

The final selection of an S-curved semi-displacement monohull provides the optimal compromise for our application. It offers the internal volume of a cargo vessel, the shallow-water

access of a flatbottom vessel, and the self-righting safety required for autonomous operation in hazardous coastal conditions.

4.2 Material Selection

The vessel's construction uses hybrid manufacturing materials designed to have a high degree of layer complexity while maintaining the structural integrity required for maritime operations.

4.2.1 Hybrid Composite Architecture

The manufacturing process went from additive manufacturing to composite reinforcement to balance precision and durability. The following materials were used to create the vessel's structure.

- **3D Printed PLA Outline:** The internal hull geometry used 3D printed using Polylactic Acid (PLA) filament. This allowed for the exact structure of the lofted S-curve semi-displacement monohull design that would be difficult to produce using traditional manual carving or molding techniques.
- **Fiberglass Composite Cloth:** While PLA provided the geometric foundation, it lacks the resistance and long-term waterproof stability needed for a functional vessel. The hull was reinforced with Fiberglass Composite skin paired with an Epoxy for strength and waterproofing.
- **Marine-Grade Epoxy:** The fiberglass layers were bonded to the PLA substrate using Marine-Grade Epoxy, which created a high-strength-to-weight ratio shell that protects the internal PLA from moisture and UV damage.

4.3 Enclosed Top Shell

The protective shell was developed in CAD as a streamlined, aerodynamic component that serves as a mounting platform for our Solar panels, Wi-Fi antenna, GPS, and Camera. The shell consists of two main components held together by a rubber latch mechanism, allowing for rapid internal access during field maintenance while maintaining a secure connection during operation.

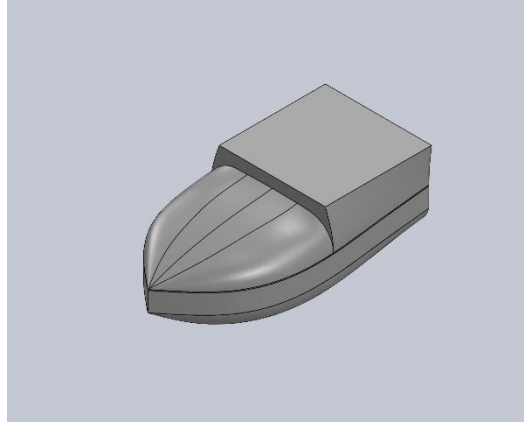


Figure 10: Isometric view of bottom hull & top shell assembly.

4.4 Waterproofing & Structural Integrity:

Structural integrity is reached through a combination of fiberglass composite construction, finished with marine-grade epoxy and a protective primer/paint layer. The waterproofing is maintained through a marine rubber sealant tape applied at the edge of the hull and top shell intersection. An additional internal protective box was created and installed to hold sensitive electronics. They are housed at the back end of the hull, with the internal enclosure providing secondary protection against any kind of water leakage. Field testing confirmed 100% waterproof protection during a 30 second submersion test and autonomous navigation tests, validating the compressional seal design.



Figure 11: Top section of hull.

5.0 Electrical Design

5.1 System Overview and Environmental Constraints

The electrical architecture of the autonomous boat craft is built upon a continuous 14.8 DC central power bus. The system is engineered to meet several objectives: maintaining a high-capacity lithium-polymer power reserve, harvesting solar energy through a two-panel array to extend operational life, and distributing sustained power to the propulsion and control systems. A large challenge of the design is the physical environment of the boat itself. Because the hull is constructed from a thick, sealed fiberglass composite, it provides excellent waterproofing but creates a major thermal challenge. Fiberglass is a very good thermal insulator, meaning any heat generated by the electrical components through resistance or high-current draw is trapped inside the hull with no way of escaping. To counter this, the entire electrical circuit is over-engineered. By using oversized wiring and high-efficiency components, we minimize the internal resistance and heat generation, protecting the craft from thermal runaway while ensuring the electronics can survive the high humidity of the marine environment.

5.2 Primary Power Storage and Chemical Management

The core energy source is a 14.8 20000mAh 4S Lithium Polymer (LiPo) battery. This battery choice was made because LiPo chemistry offers the high energy density needed for long-range autonomous missions, providing a total energy capacity of nearly 300 Watt-hours. However, lithium batteries are notoriously sensitive and require precise management to prevent permanent damage or fire. To deal with this, a 4S 30A Battery Management System (BMS) is wired directly to the battery. This BMS acts as guard for the entire boat. It uses a balance lead to monitor the voltage of each of the four cells individually and in real time. If the propulsion motors attempt to draw more than the rated 30 Amps, or if any single cell drops below the safe 3.0V limit, the BMS instantly cuts the connection. This safety feature is vital because it prevents the battery from entering a state of deep discharge or overheating, which is very dangerous inside an unventilated fiberglass hull.

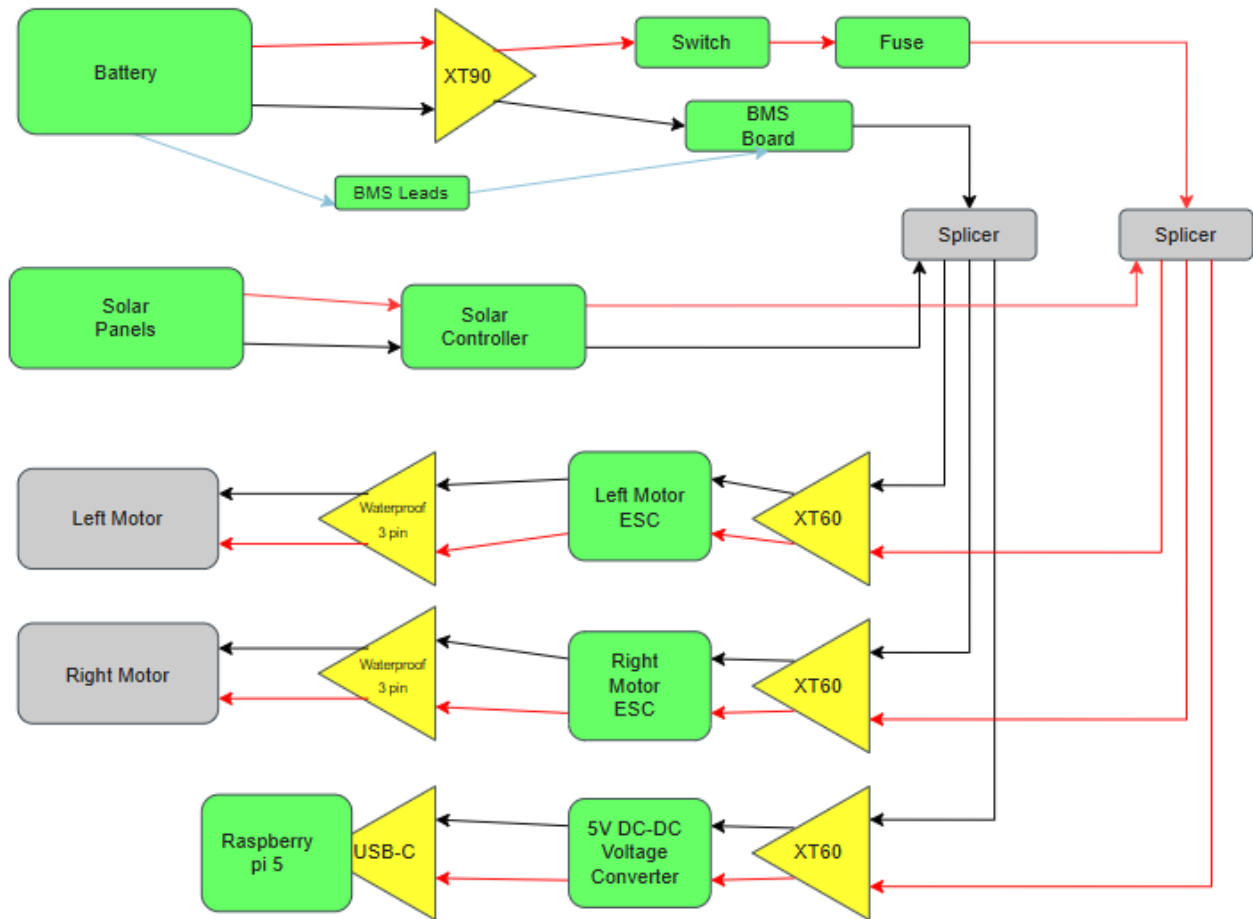


Figure 12: Circuit diagram.

5.3 Solar Power Generation and Boost Conversion

To increase the duration of autonomous missions, the craft uses two 2W 12V solar panels mounted at the top. These panels are wired in parallel to maximize current collection. A major technical gap with that set up is that a 12V solar panel cannot naturally charge a 14.8V battery because electrical current only flows from a higher voltage to a lower one. To bridge this gap, the system utilizes an MPT-7210A Digital Solar Charge Controller. This acts as a specialized DC-to-DC boost converter. It takes the incoming 12V from the sun and uses high-frequency switching and inductors to step up the voltage to a constant 16.8V, which is the full-charge voltage of the 4S battery. This boosted power is fed into the main distribution blocks, allowing the solar energy to either help power the motors during the day or flow backward through the BMS to trickle-charge the battery which the boat is stationary.

5.4 Transmission Medium and Galvanic Protection

Transferring high current throughout the boat requires a robust wiring system to prevent voltage drops and heat buildup. All primary power lines are constructed using 10 AWG stranded silicone wire. We chose 10-gauge copper because its thickness allows it to carry well over the 30A limit of the BMS without the warming up, which is a necessary precaution inside the insulated hull. The silicone wrapping and high strand count make the wire extremely flexible, ensuring that the constant vibrations from the twin 300W motors do not cause the metal to fatigue or snap over time. Furthermore, the marine environment is highly corrosive. To prevent galvanic corrosion, where different metals eat each other away in the presence of moisture and electricity, every connection point used tinned copper ring terminals. These terminals are then sealed with heat shrink or electrical tape to ensure that no salt air or moisture can reach the copper, maintaining a low-resistance connection for the life of the craft.

5.5 Safety Failsafes and Manual Overrides

Beyond the digital protection provided by the BMS, the boat features two layers of physical failsafes to protect the hardware from catastrophic failure. The first is a waterproof inline MAXI fuse holder equipped with a 40-Amp blade fuse. This fuse serves as a secondary line of defense. If a major short circuit occurs and the solid-state BMS fails to trip, the fuse will physically melt and break the circuit. We chose a 40A fuse to provide a small buffer over the 30A BMS limit, allowing for very brief surges in motor torque without causing nuisance trips. The second safety layer is a heavy-duty marine battery disconnect switch. This switch is mounted in an accessible location and allows an operator to manually kill all power to the grid instantly. This is essential during transport, maintenance, or in the event of a software glitch where the autonomous system fails to respond to commands.

5.6 Propulsion Sub-Circuit and Current Limiting

The propulsion system consists of two 16V 300W motors, which represent the largest electrical load on the boat. These motors are controlled by two independent 20A Electronics Speed Controllers (ESCs). Because the combined power of these motors could theoretically exceed the 30A limit of the battery management system, the ESCs are programmed to act as a bottleneck. By limiting each motor to a maximum of 15 to 20 Amps, we ensure that even at full throttle, the motors will not trigger a total system shutdown by the BMS. Additionally, the ESCs play a major part in electrical health by acting as a buffer. High-power brushless motors create a significant amount of electrical noise and voltage ripples as they spin. The ESCs help contain this noisy power within the propulsion loop, preventing high-frequency inference from traveling back through the power lines and crashing the navigation computer.

5.7 Logic and Navigation Sub-Circuit Stability

The brain of the autonomous craft is a Raspberry Pi 5, which handles all the GPS processing, sensor data, and steering logic. Unlike the motors, which can handle slight

fluctuations in voltage, the Raspberry Pi is a sensitive piece of microelectronics that requires a perfect 5V input to function. To provide this, the system uses a high-amperage DC-DC Buck Converter. This converter takes the fluctuating 14.8V from the main battery bus and steps it down to a stable 5V. It also acts as a power filter, when the propulsion motors start or stop suddenly, they can cause the system voltage to sag or spike. The Buck Converter smooths out these fluctuations, ensuring the Pi 5 never experiences a brownout. Without this dedicated regulation, the Pi would likely reboot every time the boat attempted a sharp turn or burst of speed, leading to a total loss of autonomous control.

6.0 Software & Autonomy

The software and autonomy system of the autonomous boat craft is built on a Robot Operating System 2 (ROS 2 Humble) framework running on a Raspberry Pi 5 single-board computer. The architecture is designed to execute fully autonomous waypoint navigation missions without real-time human intervention while providing manual override capability for operator safety. The autonomy stack was developed on top of an existing open-source ROS 2 codebase from a prior WPI MQP project (Mann et al., 2025), with significant new nodes and launch infrastructure written for this application. This section describes the system architecture, navigation stack, sensor integration, computer vision pipeline, safety systems, and operator interface.

6.1 System Architecture Overview

6.1.1 Codebase Foundation & Prior Work

The autonomy stack for this project was developed on top of the open-source ROS 2 repository produced by the 2024-2025 WPI MQP project *The Design and Prototyping of a Low-Cost & Efficient Ocean Cleanup Robot* (Mann et al., 2025). That codebase provided a functional set of ROS 2 sensor driver nodes, a motor controller interface, a YOLO-based object detection pipeline, and a keyboard teleoperation node, all validated on a prior autonomous maritime platform. Reusing this foundation avoided redundant low-level driver development and allowed this project's software effort to focus on the higher-level autonomy, mission planning, and safety functionality required for emergency delivery operations.

The following nodes were inherited from the prior repository and reused without modification:

- `gps_pub` - GPS receiver driver; reads position from the `gpsd` daemon and publishes `sensor_msgs/NavSatFix` on `/get_GPS`.
- `imu_pub` - IMU and magnetometer driver; reads the BerryIMU v3 (LSM6DSL accelerometer/gyroscope and LIS3MDL magnetometer) over I2C and publishes `sensor_msgs/Imu` on `/IMU_data`.
- `water_sensor` - Hull leak detection driver; monitors GPIO pin 27 via `gpiozero` and publishes `std_msgs/Bool` on `/emergency_stop` upon water detection, then issues a system shutdown command.
- `object_detector` - YOLO inference node; opens the camera via OpenCV, runs YOLOv8n inference, publishes `std_msgs/Bool` on `/obstacle_detected` and a compressed camera stream on `/camera/image/compressed`.

- `kalman` - Extended Kalman Filter state estimation node; fuses `/IMU_data`, `/get_GPS`, and `/set_motor_speeds` into a filtered state vector published on `/get_state`.
- `motor_control` - Low-level thruster command interface; subscribes to `/set_motor_speeds` and forwards signed speed commands to the ESP32 microcontroller over UART serial at 115200 baud on `/dev/ttyUSB0`.
- `teleop` - Keyboard teleoperation node for manual override over SSH; uses raw terminal I/O via `tty/termios` and publishes motor commands on `/set_motor_speeds`.
- `mjpeg_stream` - Auxiliary video streaming node; subscribes to `/camera/image/compressed` and serves a live MJPEG stream over HTTP on port 8080 for operator monitoring.

The following nodes and configuration files were written from scratch for this project:

- `waypoint_nav.py` - Autonomous waypoint navigation controller; subscribes to `/get_GPS` and publishes differential motor commands on `/set_motor_speeds` at 10 Hz to execute sequential GPS-based missions.
- `obstacle_avoidance.py` - Motor command gating node; subscribes to both `/obstacle_detected` and `/set_motor_speeds` and republishes motor commands as `/set_motor_speeds_safe` after applying a three-state obstacle response finite state machine.

The division of inherited and original contributions is consistent with standard iterative robotics development practice, where proven low-level infrastructure is preserved and extended rather than rebuilt. The prior repository is publicly available at <https://github.com/jhkeselman/cleaningoceanplastics/releases/tag/1.0> and the associated project report is archived at https://digital.wpi.edu/concern/student_works/47429f465.

6.1.2 Node Graph & Functional Layers

The full autonomy software is organized as a modular ROS 2 node graph in which each functional subsystem is encapsulated as an independent node communicating over ROS 2 topics and services. No ROS 2 services are used anywhere in the stack. All inter-node communication follows the publisher/subscriber model. This decoupled design allows individual nodes to be developed, tested, and replaced in isolation without disrupting the broader system. The Raspberry Pi 5 serves as the sole onboard computer platform, hosting all ROS 2 nodes, sensor drivers, and the computer vision pipeline simultaneously.

The active mission stack is organized as two ROS 2 packages: `sensors` and `navigation`. The `sensors` package contains `water_sensor`, `imu_pub`, `gps_pub`, `object_detector`, and `Kalman`. The `navigation` package contains `waypoint_nav`,

obstacle_avoidance, motor_control, teleop, and mjpeg_stream. A full autonomous mission is launched via `mission_launch.py`, which includes `sensors.launch.py` (bringing up `imu_pub`, `gps_pub`, `water_sensor`, and `object_detector`) and additionally launches `motor_control`, `waypoint_nav`, and `obstacle_avoidance`. The `teleop` and `mjpeg_stream` nodes are launched manually on demand.

One cross-cutting architectural feature is the `/emergency_stop` topic. The `water_sensor` node is its sole publisher. Every other active node in the stack subscribes to it. Upon receiving a `True` value on this topic, each node halts its operation. This ensures that a hull leak event produces an immediate, system-wide shutdown without requiring any central coordinator.

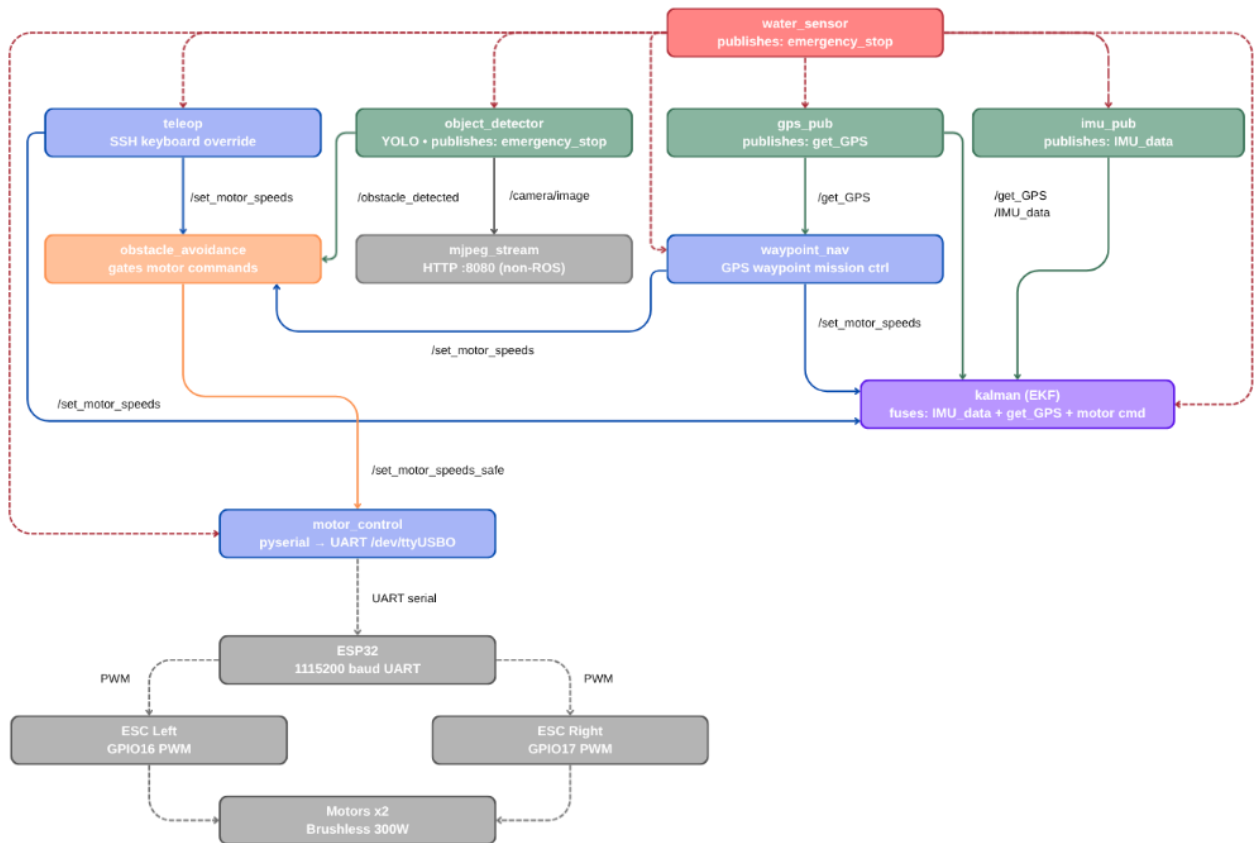


Figure 13: ROS 2 node graph showing all active nodes, topic connections, and the `emergency_stop` broadcast.

6.2 Navigation Stack

Autonomous navigation is implemented as a proportional waypoint-following mission controller in `waypoint_nav.py`. Prior to deployment, a mission is defined as an ordered sequence of GPS coordinates hardcoded as latitude/longitude pairs in the node source file. At 10 Hz, the navigation node reads the latest GPS fix from `/get_GPS`, computes the bearing and

Haversine distance to the active waypoint, and publishes differential motor speed commands on `/set_motor_speeds`.

6.2.1 Waypoint Logic

The bearing to the target waypoint is computed using the standard spherical forward azimuth formula and normalized to the range $(-180^\circ, 180^\circ]$. A proportional turn speed is derived by multiplying the bearing error by a gain constant `KP_BEARING = 0.008`. A heading factor, defined as $\max(0, 1 - |\text{bearing_error}| / 90)$, scales forward drive to zero when the bearing error reaches or exceeds 90° , preventing the vessel from advancing while executing large-angle turns. Left and right motor commands are computed as `forward_speed ± turn_speed` and clamped to a maximum magnitude of 0.5. These are published as a `Float32MultiArray` on `/set_motor_speeds`.

The vessel advances to the next waypoint once its Haversine distance to the current target falls below an arrival radius of 2.0 meters. Upon completing all waypoints, the node publishes zero motor speeds and cancels its control timer.

It is important to note current architectural limitations of the navigation implementation. `waypoint_nav` computes an absolute bearing to the target but does not subscribe to any heading or compass topic. The controller therefore has no direct knowledge of which direction the vessel is currently facing and relies entirely on the proportional feedback loop to drive bearing error toward zero: this simplification is functional in low-disturbance conditions.

6.2.2 Sensor Fusion & State Estimation

The Kalman node implements an Extended Kalman Filter that fuses `/IMU_data`, `/get_GPS`, and the current `/set_motor_speeds` command into a continuous filtered state estimate published on `/get_state`. The IMU provides linear accelerations and angular velocity from the BerryIMU v3's LSM6DSL, while GPS provides absolute position fixes. Including motor commands in the filter inputs allows the EKF to incorporate a simplified motion model, improving state continuity during momentary GPS outages.

It should be noted that in the current active mission stack, no node subscribes to `/get_state`. The `waypoint_nav` node navigates directly from raw `/get_GPS` fixes rather than from the EKF output. The Kalman-filtered state estimate is not yet integrated into the navigation controller.

6.3 Computer Vision Pipeline

The vessel is equipped with a forward-facing USB camera that feeds a real-time object detection pipeline implemented in `object_detector`. The node opens the camera device via OpenCV, runs frame-by-frame inference using a YOLOv8n model loaded from `yolov8n.pt`, and publishes two outputs: a `std_msgs/Bool` on `/obstacle_detected`

indicating the presence of any detected object above confidence threshold, and a compressed camera stream on `/camera/image/compressed` consumed by `mjpeg_stream` for live operator monitoring. Detected video sessions are also written to `.mp4` files on the Raspberry Pi for post-mission review.

6.3.1 YOLO Object Detection

The team selected YOLOv8n as a lightweight single-stage detector suitable for interference on the Raspberry Pi 5 CPU. The model detects and classifies objects in the vessel's forward field of view in real time. The node publishes a Boolean obstacle flag rather than full bounding box data, which simplifies the downstream interface with the obstacle avoidance node. Frame resolution and interference frequency are configurable parameters subject to ongoing performance testing to balance detection latency against CPU load.

6.3.2 Obstacle Avoidance Integration

Obstacle response is handled by the `obstacle_avoidance` node, which implements a three-state finite state machine. In the `CLEAR` state, the node publishes nothing, allowing motor commands from `waypoint_nav` or `teleop` to reach `motor_control` unmodified. On a rising edge of `/obstacle_detected`, the node transitions to the `STOP` state and publishes zero motor speeds on `/set_motor_speeds_safe` for a fixed duration of 2.0 seconds. It then transitions to the `TURN` state, publishing a fixed pivot-right command `[0.2, -0.2]` until `/obstacle_detected` returns `False`, at which point it returns to `CLEAR`.

The gating mechanism relies on the fact that `motor_control` subscribes to `/set_motor_speeds_safe`, which is only published by `obstacle_avoidance`. During a `CLEAR` state, `obstacle_avoidance` publishes nothing, so the motor controller receives no commands from the safety layer. The current implementation publishes `obstacle_avoidance` commands at 10 Hz during active obstacle states, which overwrite any concurrent `waypoint_nav` output seen by downstream subscribers. This last-write-wins approach is functional but does not implement a formal message priority mechanism.

6.4 Safety Systems

Given the uncrewed nature of the vessel and the hazardous near-shore operating environment, the autonomy stack incorporates multiple layers of software and hardware safety protection.

6.4.1 Hull Leak Detection & Emergency Stop

A digital leak detection sensor connected to GPIO pin 27 is mounted inside the hull to monitor for water intrusion. Upon detecting moisture, the `water_sensor` node immediately publishes `True` on the `/emergency_stop` topic, which every other active node in the stack subscribes to and uses as an immediate halt trigger. Following the broadcast, the node issues a

`sudo shutdown -h now` system call to initiate a controlled Raspberry Pi shutdown, preventing further damage to onboard electronics. This two-stage response ensures the vessel loses propulsion and autonomous control before power is removed from the compute platform.

6.4.2 Manual Override

At any point during autonomous operation, the operator can assume direct control of the vessel via the `teleop` node over an SSH session. `teleop` uses raw terminal I/O via `tty` and `termios` to capture keyboard input without requiring a graphical interface, making it fully compatible with SSH-based remote operation. When active, `teleop` publishes motor commands on `/set_motor_speeds` at the same rate as `waypoint_nav`. Since `obstacle_avoidance` subscribes to this topic as its command input, `teleop` commands flow through the same obstacle gating pipeline as autonomous commands, preserving safety behavior during manual operation.

6.4.3 Operator Interface & Telemetry

Operator communication with the vessel is conducted entirely over a 2.4 GHz WiFi link. Through this interface, the operator can monitor live ROS topic output, activate or terminate the autonomous navigation stack via ROS 2 launch commands, assume manual keyboard teleoperation control, and review onboard logs for post-mission diagnostics. The `mjpeg_stream` node provides a live camera feed accessible via any web browser on the same network at port 8080, enabling visual situational awareness without requiring ROS tooling on the ground station. The WiFi-only communication architecture imposes a practical operational range constraint.

6.5 Implementation Status

The following functions are currently functional:

- ROS 2 Humble environment configured and running on the Raspberry Pi 5.
- GPS driver node (`gps_pub`) publishing `NavSatFix` messages from the `gpsd` daemon.
- IMU and compass driver nodes (`imu_pub`) publishing sensor data from the BerryIMU v3 over I2C.
- EKF state estimation node (`kalman`) fusing GPS, IMU, and motor command data into a filtered state vector.
- Waypoint navigation node (`waypoint_nav`) executing sequential point-to-point GPS missions in controlled conditions using a proportional bearing and distance controller.
- Obstacle avoidance node (`obstacle_avoidance`) implementing a three-state FSM that gates motor commands in response to YOLO detections.

- Hull leak detection node (`water_sensor`) with system-wide emergency stop broadcast and Pi shutdown trigger.
- Manual override via SSH keyboard teleoperation (`teleop`).
- YOLO inference pipeline (`object_detector`) publishing obstacle detection flags and live compressed video.
- Full autonomous mission launch via `mission_launch.py`.

7.0 Manufacturing & Assembly

The autonomous surface vessel was manufactured using a hybrid approach combining custom fabrication and commercial off-the-shelf (COTS) components. Emphasis was placed on structural integrity, waterproofing, and modular internal assembly to ensure reliable operation in a harsh marine environment.

7.1 Housing Fabrication

The hull of the vessel was fabricated in-house using a fiberglass composite layup process over a 3D-printed mold. The mold was designed in CAD and printed in four separate sections: two for the upper hull and two for the lower hull. Each pair of sections was bonded together to form complete top and bottom molds.

Fiberglass sheets were then applied to the mold using a hand layup process. A total of three layers of fiberglass reinforcement were used, each saturated with marine-grade epoxy resin to create a rigid composite structure. This layering process was selected to balance structural strength with overall weight constraints. After curing, the composite hull sections were removed from the molds and prepared for assembly.



Figure 14: Sanding and smoothing of hull layers.

The final hull geometry follows a semi-displacement monohull design, providing sufficient buoyancy and stability for payload transport while maintaining hydrodynamic efficiency.

7.2 Structural Assembly

Following fabrication, the upper and lower hull sections were aligned and joined to form the complete vessel body. Mechanical fastening and adhesive bonding were used to ensure structural rigidity. To enable repeated access to internal components while maintaining watertight integrity, the hull interface was designed as a sealed enclosure rather than a permanently bonded joint. This allowed the system to remain serviceable throughout testing and iteration.

7.3 Waterproofing and Sealing

Given the operating environment, waterproofing was a critical design requirement. A continuous rubber seal was applied along the perimeter of the lower hull interface. This seal was adhered using marine-grade adhesive to ensure consistent compression and long-term durability.

The upper and lower hull sections were secured using eight external rubber latches distributed evenly around the perimeter. These latches apply uniform compressive force on the seal, preventing water ingress during operation, including partial submersion and wave impact.

All cable entry points, including those for propulsion wiring, solar panels, and communication systems, were routed through drilled openings in the hull. These interfaces were sealed to prevent leakage and maintain enclosure integrity.



Figure 15: Latch installation. & Figure 16: Completed hull parts.

7.4 Internal Component Housing

To organize and protect onboard electronics, a custom two-part internal housing was designed and fabricated using 3D printing. The lower section of the housing serves as a mounting platform for all electronic components, including the battery, power distribution system, and control hardware. The upper section acts as a protective cover, shielding components from moisture and mechanical disturbance.

This modular housing approach provides several advantages:

- Secure mounting of sensitive electronics
- Simplified assembly and maintenance
- Isolation of components from hull vibrations and incidental water exposure

The Raspberry Pi 5, Battery Management System (BMS), DC-DC converters, and associated wiring were all integrated within this enclosure.

7.5 Propulsion System Integration

The propulsion system consists of two electric motors mounted directly to the lower hull. Each motor was mechanically fastened by drilling mounting points through the hull structure, ensuring a rigid and stable connection. Care was taken to align the motors symmetrically to enable effective differential thrust control for steering. Electrical connections from the motors were routed through sealed hull penetrations to the internal power and control systems.

7.6 Electrical and Sensor Integration

Once the structural and mechanical components were assembled, the internal electronics were installed within the housing. The power system, including the 14.8V LiPo battery, BMS, and solar charge controller, was wired according to the electrical architecture described in Section 5.

External components were then integrated:

- GPS module: Mounted within the housing for navigation
- Solar panels: Installed on the top surface of the hull to maximize solar exposure
- Wi-Fi antenna: Mounted externally to maintain communication range

All external components were connected to the internal system via sealed cable routing paths to preserve waterproofing.

7.7 Assembly Process Overview

The full manufacturing and assembly process followed a structured sequence:

1. 3D printing of hull molds and internal housing components
2. Fiberglass layup with epoxy resin (three layers)

3. Curing and removal of composite hull sections
4. Bonding and alignment of hull components
5. Installation of rubber seal along hull interface
6. Mounting of external latching system
7. Installation of internal electronics housing
8. Integration of electrical components and wiring
9. Mounting of propulsion motors
10. Installation of external systems (solar panels, antenna, sensors)
11. Final sealing and system verification

7.8 Manufacturing Considerations and Cost

The vessel was developed under a fixed project budget of \$1,250, which was fully allocated toward materials, electronics, and fabrication resources. The use of in-house manufacturing techniques, such as 3D printing and fiberglass layup, significantly reduced production costs while allowing full customization of the design. The reliance on COTS electrical components further simplified integration and reduced development time. However, the fabrication process required careful attention to sealing, alignment, and structural reinforcement to ensure reliability in marine conditions.

7.9 Summary

The manufacturing and assembly process successfully produced a lightweight, structurally robust, and waterproof autonomous vessel. The combination of fiberglass composite construction, modular internal design, and sealed enclosure enabled effective integration of propulsion, power, and autonomy systems within a compact platform suitable for near-shore emergency delivery applications.



Figure 17: Finished craft hull.

8.0 Prototyping & Testing

The prototyping and testing phase was used to evaluate whether the completed autonomous boat craft satisfied the system-level design specifications established earlier in the report. Since this project was developed as a proof-of-concept, the purpose of testing was not to certify the vessel for real disaster response, but to determine whether the prototype could successfully demonstrate its main engineering functions in simple water conditions.

The most important requirements tested were flotation, payload support, water resistance, forward propulsion, differential steering, autonomous waypoint navigation, sensor integration, and solar subsystem integration. These tests were organized around the behavior of the full vessel rather than only individual components. This was important because the performance of the boat was based on how the hull, payload, propulsion system, sensors, and internal layout worked together.

For example, buoyancy could not be fully evaluated from the hull geometry alone because the final weight, payload placement, and internal component layout all affected trim and stability. As well as propulsion and steering had to be tested after assembly because the motors, hull resistance, and weight distribution all influenced how the craft moved in water. The testing process therefore focused on verifying the completed prototype as an integrated autonomous surface vessel.

Test	Requirement	Method	Result	Pass/Fail
Flotation	Support 25.3 lb loaded mass	Static water test	Maintained positive flotation	Pass
Leak test	No visible ingress	30 sec submersion	No internal water ingress	Pass
Payload test	Carry 6.0 lb	Loaded flotation test	Stable trim	Pass
Navigation	Reach waypoint within target radius	GPS waypoint mission	3 successful mission cycles	Pass
Solar integration	Provide charging voltage	Solar panel/MPPT test	Add actual voltage/current	Pass/Partial
BMS/current test	Sustain propulsion load	Full throttle/startup	30A BMS failed	Fail

Table 4: Testing evaluation table.

8.1 Flotation and Payload Testing

The first major test verified whether the vessel could remain positively buoyant under its expected operating condition. The fully assembled boat weighed 19.3 lb without payload. With the rated 6.0 lb emergency payload added, the loaded operating weight was approximately 25.3

lb. This value was compared against the calculated maximum buoyant support of 34.4 lb, leaving approximately 9.1 lb of remaining buoyancy in the loaded condition.

During flotation testing, the boat was placed in water and observed for static trim, freeboard, and overall stability. The goal was to confirm that the hull did not sit too low in the water and that the payload did not cause excessive leaning, bow-down trim, or stern-down trim. This test directly supported the buoyancy and payload specifications by showing that the vessel could support both its internal systems and the design payload while maintaining positive flotation.

This test was also useful because it showed whether the internal layout and payload placement were good enough for the application. Since the craft is intended to operate autonomously, stable trim is important for propulsion, steering, and sensor orientation. A boat that floats but sits unevenly would be difficult to control. The successful flotation result confirmed that the hull displacement and weight distribution were suitable for continued water testing.

8.2 Leak and Waterproofing Test

After confirming flotation, the next test evaluated the waterproofing of the hull and access structure. The boat was exposed to controlled water conditions during a short-duration leak and submersion test, approximately 30 seconds of full submergence with zero leakage. The purpose of this test was to check whether water entered through the removable top section, wire pass-throughs, sealed joints, or any other potential leak paths.

This test was especially important because the boat contains internal systems and payload space that must remain protected during operation. Before and after the test, the vessel was inspected for signs of water ingress. Attention was given to the access seam, sealed wire exits, and areas around externally mounted components. We noticed initial potential for leakage on the left corner outside seal, which we addressed with more marine grade silicon to plug the gap.

The leak test also helped validate the usage requirement from the design specifications. The boat needed to be accessible for payload loading and maintenance, but it also needed to reseal effectively after being opened. Passing this test showed that the removable access concept could support controlled prototype testing without compromising the internal environment.

8.3 Propulsion and Steering Testing

Propulsion and steering tests were completed to verify that the loaded vessel could move through water and maneuver using differential thrust. The craft was designed with two electric thrusters, allowing it to move forward when both motors produced similar thrust and turn when the left and right motors produced different thrust levels.

The propulsion test confirmed that the vessel could move forward under its assembled weight and payload condition. The steering test then evaluated whether the boat could rotate and

change direction by varying thrust between the two motors. This was important because the project required low-speed maneuverability rather than only straight-line motion. In a near-shore delivery scenario, the craft would need to correct its heading, approach a target area, and maneuver in constrained water conditions.

These tests also showed why propulsion could not be evaluated only by motor rating. A motor that produces enough thrust on its own still must work with the actual hull shape, weight distribution, and payload condition. The successful propulsion and steering tests demonstrated that the assembled prototype could move and turn in a controlled manner.

8.4 BMS Failure

During the second test of the fully assembled craft, the 30A battery management system experienced a catastrophic thermal failure due to an underspec'd continuous current rating. As the 2-foot autonomous boat craft transitioned from a dead stop to forward momentum, the surrounding water provided immense immediate resistance against the stationary propellers. This physical inertia forced the dual 16V 300W motors to pull an instantaneous startup surge estimated between 80 and 100 amps, a spike that far exceeded the maximum burst limit of the onboard safety circuit.

Once the logic system commanded the throttle to open fully, the combined 600-watt propulsion load demanded a sustained draw of over 40 amps from the 4S lithium-ion power plant. Because the 30A board was mathematically incapable of cleanly passing that volume of electrical flow, the internal MOSFETs bottlenecked the system and acted as a massive resistor. The excess trapped current was rapidly converted into extreme heat, which operated entirely outside the engineered thermal limits of the circuit. Within moments, the solid-state switches physically melted and permanently severed the primary power connection, leaving the fiberglass hull completely dead in the water and severing all propulsion capabilities.

8.5 Autonomous Navigation Testing

With floatation, waterproofing, and propulsion validated, the team tested the system for autonomous waypoint navigation. The goal was to confirm that the full software stack could execute a GPS-based mission from launch to completion without operator input. The team defined the mission as a sequence of hardcoded latitude/longitude waypoints loaded into `waypoint_nav.py` prior to deployment. The full autonomous stack was brought up using `mission_launch.py`, which initialized the sensor drivers, YOLO pipeline, waypoint navigation controller, and obstacle avoidance node simultaneously.

During testing, the vessel successfully received a GPS fix, computed bearing and distance to the first target, and began issuing differential motor commands to steer toward it. The proportional controller reduced bearing error over successive control cycles, and the vessel advanced through the waypoint sequence. It registered arrival once Haversine distance fell below

the 2.0-meter acceptance radius. The team completed three successful mission cycles in a controlled aquatic environment, satisfying the proof-of-concept navigation objective.

The team evaluated navigation accuracy by observing the vessel's final position relative to each target waypoint. Results confirmed repeatable point-to-point delivery within the design tolerance of approximately 3 meters over a 100-meter mission. The controller's heading factor, which scales forward drive toward zero during large bearing errors, was effective in preventing the vessel from advancing during sharp turns, keeping the trajectory reasonably clean without compass feedback.

Limitations observed during this testing are consistent with those identified in Section 6. Since `waypoint_nav` computes absolute bearing without a heading subscription, course correction depended entirely on GPS feedback latency and proportional gain tuning. In low-disturbance conditions this was sufficient, but in open water with current or wind disturbance, this limitation is likely to become even more significant. Validation under those conditions is a target for future practice.

8.6 Sensor Integration Testing

The team tested all active sensor nodes individually and in combination during mission runs. The GPS drive consistently published fix data through the `gpsd` daemon throughout each test. The IMU drive published orientation and acceleration data continuously over I2C without dropout. The YOLO interference pipeline maintained real-time detection output across all sessions, with the `object_detector` node publishing to `/obstacle_detected` and streaming compressed frames to `mjpeg_stream` on port 8080 simultaneously. The operator successfully monitored the live camera feed from a laptop browser on the same WiFi network throughout each run, confirming the telemetry architecture functioned properly.

The hull leak detection node was verified separately during the waterproofing test. Upon manual triggering, the `water_sensor` node published `True` on `/emergency_stop`, making subscribed nodes to halt as expected, and initialing the system shutdown sequence. This confirmed the safety architecture described in Section 6.4.1 behaved correctly in practice.

8.7 Solar Integration Testing

The team tested the two 12V 3W solar panels and the MPT-7210A boost charge controller to confirm that solar energy was being harvested and fed into the main power bus during the vessel's operation. Under direct sunlight conditions, the charge controller stepped up the panel output voltage to 16.8V and delivered supplemental current to the distribution bus. The team confirmed that the panels offset a measurable portion of the system's power draw during daylight operation, validating the solar integration objective at the prototype level. The team notes that quantitative endurance testing under sustained solar conditions is a target for future outdoor field validation.

9. Broader Impacts

This chapter reflects on the broader implications of this MQP project across five core pillars new engineers must learn: social, global and economic impacts, engineering ethics, environmental concerns, codes and standards. Although this project is a prototype, these reflections consider the full-scale realization of the system as a humanitarian aid platform.

9.1 Engineering Ethics

The ASME Code of Engineering Ethics holds engineers to a standard of public safety, honesty, and competence (ASME, 2012). ASME guides engineers to successful and reliable designs throughout all industries, from EPC (Engineering, Procurement, Construction) to autonomous robotics. Several of its core principles directly shaped the design decisions made throughout this project.

The obligation to hold public safety above all else was the primary driver behind the vessel's autonomous design. By removing human operators from debris-filled and contaminated near-shore disaster environments, the system directly reduces the risk of injury or death to rescue personnel (Gavalas et al., 2026; Jorge et al., 2019). The team implemented every major safety subsystem in direct accordance with this obligation. Examples of this include: the hull leak detection emergency stop, the Battery Management System's overcurrent and cell undervoltage protection, the 40A physical fuse, and the manual override capability.

The team reflected honesty in representation through its transparency about the system's current state. The report explicitly documents what has been validated versus what remains under development, rather than overstating the present potential. This is particularly important given the humanitarian deployment context, where stretched capability claims could lead to false reliance during an actual emergency (ASME, 2012). This principle also informed the team's explicit attribution of all inherited software notes to the prior WPI MQP project (Morris et al., 2025), clearly delineating original contributions.

Looking ahead, one ethical aspect is the vessel's autonomous decision making when detecting people in the water. Current YOLO-based obstacle detection represents a first step but edge cases, such as when the fastest delivery passes a person, requires carefully defined behavioral policies and extensive field validation before real world deployment (IMO, 2021).

9.2 Societal & Global Impact

The primary beneficiaries are coastal disaster-affected populations, particularly in regions where poor infrastructure is fragile and humanitarian logistics networks are limited. Events such as Typhoon Haiyan, the 2010 Haiti earthquake, and Cyclone Sidr each demonstrated the same failure mode: supplies cannot reach populations because near shore last mile delivery was blocked by destroyed docks and inaccessible coastlines (Yoho et al., 2013; Gavalas et al., 2026). For humanitarian organizations such as the Red Cross, FEMA, or the US Navy, this system

reduces crew exposure during the most dangerous phase of disaster response by shifting operators to supervisory roles (Jorge et al., 2019). Its solar-powered architecture further enables operation where fuel resupply chains have collapsed. This is precisely the conditions present after a major coastal disaster.

Unintended consequences include potential interference with manned rescue operations if collision avoidance procedure fails, and the reduction of human support in aid delivery, which also carries cultural importance in many communities. Territory related questions applying to regulations in international waters also remain unresolved, these factors reinforced the need for robust human oversight and clear operational protocols prior to any development (IMO, 2021). While jurisdictional ambiguity is already a current issue for open ocean related aid service, our design works compliantly among existing support systems.

9.3 Environmental Impact

By replacing combustion fueled rescue boats with an electrically powered solar vessel, the system eliminates hydrocarbon emissions and the associated risk of fuel spills for ecologically protected near shore areas (Manley, 2008). Electric propulsion also produces significantly less acoustic noise, reducing disturbances to marine life sensitive to low-frequency noise.

The environmental cost of the LiPo battery, including lithium mining, manufacturing, and end of life disposal, must be considered in a full life cycle assessment. Lithium extraction has been shown to have impacts such as water consumption and habitat disruption (Nitta et al., 2015), and responsible disposable protocols would be needed. The fiberglass composite hull is also non-biodegradable and would need appropriate end of life handling. Finally, an autonomous vessel operating without a human observer may interact with marine wildlife in ways a manned vessel would avoid, where any future iterations to the detection system would explicitly account for this.

9.4 Codes & Standards

IMO Guidelines for Maritime Autonomous Surface Ships (MASS, 2021) establish the regulatory framework for autonomous vessel operation including degrees of autonomy and safety management (IMO, 2021). A deployed version of this system would additionally be subject to COLREGs (1972), which mandates proper lookout, safe speeds, and collision avoidance for all vessels underway (IMO, 1972). The current obstacle detection program represents a first step toward COLREGs compliance, but full behavioral compliance would require more in depth situational awareness than the present prototype provides, including advanced hardware and software that lie outside of the scope of this project.

On the electrical side, IEC 62133 is the standard for lithium battery safety including overcharge and short-circuit protection (IEC, 2017), which the BMS and repeat fuse address directly. ABYC E-11 specifies marine wiring and circuit protection practices (ABYC, 2018),

with which the team's use of 10 AWG marine-grade silicone wire, tinned copper-terminals, and a marine disconnect switch aligns with. External connectors and cable glands are rated to IEC 60529 IP68 for continuous submersion protection (IEC, 2013).

9.5 Economic Factors

At the prototype level, the use of commercial off-the-shelf components and open-source software (Morris et al., 2025) keeps the electronics and electrical subsystem cost in the hundreds of dollars. It sits well below commercial ASV platforms such as the SeaRobotics SR-Surveyor or Open Ocean Robotics Data Xplorer, which are priced in tens to hundreds of thousands of dollars (SeaRobotics, 2025; Open Ocean Robotics, 2026).

A production deployment unit, incorporating marine compliant electronics, repeat communication hardware, payload mechanisms, and regulatory compliance testing, is estimated at \$5,000-\$15,000 per vessel. This is economically competitive with a single crewed rescue boat deployment when fuel, crew, insurance, and maintenance are factored in (U.S. Coast Guard, 2022). The elimination of fuel resupply requirements provides additional recurring cost savings in disaster environments where supply chains are disrupted (Hine et al., 2009; Bertram, 2012). Primary procurement stakeholders, such as FEMA, naval humanitarian units, and NGOs such as the Red Cross, routinely budget for specialized logistics assets, and this platform's unit cost represents a modest addition relative to its operational capability (Van Wassenhove, 2006). One acknowledged trade-off is the WiFi-only communication architecture: upgrading to radio telemetry or cellular would increase unit cost but substantially expand operational range, a decision that end user agencies would assess depending on their mission requirements.

10.0 Conclusions & Future Work

The autonomous boat craft for emergency naval delivery was designed, fabricated, and tested as a proof-of-concept prototype demonstrating that a compact, solar-assisted autonomous surface vessel can perform waypoint-based coastal delivery missions without real-time human intervention. The system successfully met its core engineering objectives. The fiberglass composite hull maintained positive buoyancy at 56.1% displacement utilization unloaded and approximately 73.5% under the 6.0 lb design payload, with a calculated draft of 3.31 inches appropriate for shallow near-shore operation. The electrical architecture, centered on a 14.8V 20,000 mAh LiPo battery with BMS protection and solar supplementation, sustained all onboard systems through controlled test sessions. The autonomy stack executed three successful GPS waypoint missions in controlled water conditions, achieving delivery accuracy within approximately 3 meters over a 100-meter course.

Not all objectives were fully completed at the time of submission. The BMS thermal failure discussed in Section 8 revealed a major issue between the startup current drawn by the propulsion system and the current rating of the protection circuit. In addition, open-water testing under realistic near-shore conditions, including waves, wind, and water currents, has not yet been fully completed. On the software side, the `waypoint_nav` controller currently navigates using only raw GPS data and does not yet use compass feedback. Similarly, although the Extended Kalman Filter (EKF) node generates a filtered state estimate, that data is not currently being used by the active navigation system. These limitations represent some of the main areas for future improvement.

10.1 Proposed Design Improvements

Future iterations should transition from the current flat-sided hull geometry to a specialized hull form, such as a deep-V or plumb bow design. While the current semi-displacement monohull met the basic buoyancy requirements, a deep-V entry would allow the vessel to "slice" through chop more effectively, reducing the slamming loads on the fiberglass hull during near-shore delivery. This would improve the stability of the onboard camera feed and GPS lock by reducing high-frequency pitch and roll oscillations in unsettled water.

To address the high displacement utilization under the 6.0 lb. payload, a longitudinal volume redistribution is recommended. By widening the hull specifically at the midship and stern, the center of buoyancy would shift to better support the heavy 14.8V battery and upgraded 100A BMS. This would allow the vessel to maintain a level trim at the design draft of 3.31 inches, preventing the "stern-squatting" effect that occurs when the propulsion motors are at full thrust, which currently risks water overtopping the rear latching interface.

Finally, the vessel should incorporate integrated spray rails and a flared bow. Since the 2-foot hull operates with limited freeboard, molded spray rails would deflect water away from the upper hull seam and solar panels during forward motion. This mechanical "dry-deck" strategy

would reduce the reliance on the rubber sealant tape alone and keep the solar panels clear of salt crusting, ensuring the two 12V 3W panels maintain peak efficiency throughout the mission duration.

10.2 Proposed Mechanical Improvements

Mechanical improvements would focus on increasing the vessel's operational safety margins and stability. While the current S-curve hull provides excellent secondary stability, a small vertical increase of the hull is recommended to increase the freeboard at the 6.0 lb design payload. This would improve the reserve buoyancy beyond the current 26.5%, providing a greater buffer against swamping in choppy coastal swells. Additionally, the propulsion system should be fitted with some kind of thruster intake guard to protect the propellers from the marine debris common in post-disaster zones. To further improve the vessel's ability in debris-filled coastal zones, future iterations should look at the transition from propeller thrusters to integrated water jet propulsion. This modification would eliminate the risk of propeller damage while further reducing the vessel's operational draft. This upgrade would require a significant redesign of the internal volume to accommodate the pump ducting and would require a shift in the navigation logic from differential thrust to nozzle specific directions. To improve the vessel's self-righting capability, the internal layout could be optimized by utilizing a custom-molded battery tray at the lowest point of the keel, further increasing the distance between the Center of Gravity and the Center of Buoyancy to ensure rapid recovery during turtling.

10.3 Proposed Circuit Upgrades

To prevent a recurrence of this thermal failure, the primary power distribution network requires a significant upgrade to handle the demanding physical environment. The most critical improvement is replacing the undersized protection circuit with a heavy-duty 100A Battery Management System (BMS). A board rated for 100 continuous amps provides well over 100% overhead for the dual motors' 40-amp running draw. This massive buffer allows the solid-state switches to effortlessly absorb the 80-to-100-amp startup spikes without generating excess heat. Upgrading to a model with active cell balancing will also physically redistribute energy between the individual lithium cells rather than simply burning off excess voltage as heat, keeping the ambient temperature inside the fiberglass hull much lower during operation.

10.3.1 Autonomous Safety and Solar Charging

Beyond raw power handling, the logic subsystem can be upgraded to actively monitor the battery's health and prevent future blind failures. By networking a Bluetooth-enabled BMS directly to the onboard Raspberry Pi 5, the logic system can actively log real-time voltage drops and current spikes exactly as the thrusters spin up in the water. Furthermore, the upgraded common-port architecture of a heavy-duty board will allow the two 12V 3W solar panels to safely trickle-charge the cells mid-mission without bypassing the safety cutoffs. The Pi can be programmed to throttle the motors dynamically if it detects the current draw pushing

dangerously close to the battery's limits, creating an autonomous software safety loop that protects the physical hardware.

10.4 Proposed Software Improvements

The team identifies several software improvements as high priority for future development. Integrating compass heading data from `imu_pub` into `waypoint_nav` would allow the controller to compute true heading error rather than relying on bearing-only proportional feedback, which is expected to meaningfully improve course correction in conditions with lateral disturbance. Connecting the EKF output from the `kalman` node into the navigation controller would replace raw GPS navigation with a smoothed, multipath-resistant pose estimate. The current last-write-wins arbitration between `waypoint_nav` and `obstacle_avoidance` on the `/set_motor_speeds` topic should be replaced with a formal priority mechanism to guarantee that obstacle override commands cannot be overwritten by navigation commands during active avoidance states. Finally, the obstacle avoidance FSM's fixed pivot-right response is a functional starting point but does not account for obstacle position or size. A future iteration incorporating bounding box data from the YOLO pipeline would enable directionally informed avoidance maneuvers rather than a fixed-direction response.

10.5 Proposed Manufacturing Improvements

To evolve the vessel from a manual prototype to a repeatable platform, the fabrication process requires a shift toward higher-precision methods. Replacing the manual hand layup process with vacuum resin infusion is recommended to achieve a more consistent resin-to-fiber ratio, effectively reducing hull weight while eliminating the microscopic voids found in the current three-layer composite. For internal assembly, the 3D-printed electronics housing should be redesigned as a removable modular sled with integrated thermal venting. This would provide necessary cooling for the upgraded 100A BMS and Raspberry Pi 5, while allowing the entire control suite to be bench-tested or swapped rapidly. Finally, the waterproofing strategy should transition from adhesive-backed seals to captive O-ring grooves molded directly into the hull interface. This mechanical seal would ensure more reliable watertight integrity under the uniform compressive force of the external latches, especially during the repeated thermal expansion and contraction cycles of long-term maritime operation.

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Appendices

Appendix A:

The team stored the code in a GitHub repository and utilized it as the ROS 2 workspace for the Raspberry Pi 5. Releases from this repository with the finalized code can be found in the following link:

- *autonomous-boat*:

<https://github.com/ndresca/autonomous-boat/releases/tag/1.0>